

The Configuration of Space :Common Base Space Platform Embedded Thermodynamic Time Arrow for General Relativity and Quantum Field Theory | Quantum-Elastic Two-Layer Fiber Bundle Model

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Abstract:

This chapter aims to establish a unified framework that integrates general relativity (GR) and quantum field theory (QFT), aligning with observed physical phenomena. The proposed model is based on a quantum network capable of conducting electromagnetic waves with quantized energy, which must achieve spatial isotropy and Lorentz covariance. A potential approach involves the random polycrystalline stacking of discrete space elementary quanta (SEQ) at the Planck scale, forming a topologically homogeneous 3D structure with defects.

The chapter further explores how quantum spin can decouple from spatial elasticity to ensure that spin degrees of freedom are not impeded by spatial elasticity. This is crucial for maintaining the independence of quantum spin properties within the elastic framework of space.

Additionally, the integration of thermodynamic time into the evolutionary model is discussed. Time is defined through the counting process of transformations within the SEQ network, reflecting the finite and discrete nature of time due to the limited number of transformations possible within a finite universe.

The harmonization of general relativity's geometric interpretation with quantum field theory's quantum spin and resonance under space deformation will be further elaborated in the following chapters. Specifically, the modulation of time under spatial deformation will be explored in detail, offering a foundational bridge between the geometric perspective of general relativity and the quantum description of physical phenomena.

Likewise, the generation of gravity by mass, where quantum field theory provides insights into the origin of mass and general relativity interprets gravity through the geometric deformation of spacetime, will be discussed in subsequent sections. These discussions aim to lay the groundwork for constructing a spatial structure platform that can seamlessly incorporate both general relativity (GR) and quantum field theory (QFT), ultimately contributing to a unified understanding of the cosmos.

Configuration of Space and Basic sets:

1.1 The universe is expanding.

1.2 The universe operates under the Law of Energy Conservation and the Law of Entropy Increase.

1.3 The universe consists of Space Elementary Quanta (SEQ), discrete Planck-scale units, forming a topologically homeomorphic 3D structure (adjacency relations remain preserved while individual SEQ energy states may vary). Importantly, this polycrystalline structure necessarily contains primordial Periodic or Random Topological Dislocation to ensure physical isotropy, yet maintains strict 3D topological homeomorphism as these defects are cosmologically frozen and adjacency-preserving since the birth of the universe. The distortion of light around black holes demonstrates that gravitational and electromagnetic field quanta are coupled, suggesting they originate from the same quantum field in different excited states—leading to the SEQ hypothesis. The spacing and tension between adjacent SEQ can be modulated by gravitational or equivalent gravitational fields.

1.4 All field quanta and elementary particles represent different energy excitation states of SEQ, expressible as 3D dynamic structural matrices of SEQ.

1.5 SEQ possess a ground state energy (e.g., ground-state spin or vibrational modes). If ground-state spin chirality is fixed, this may explain parity violation. The ground-state energy of SEQ could also account for the cosmological constant in General Relativity. This framework shows strong alignment with Loop Quantum Gravity theory.

1.6 Adjacent SEQ maintain a dynamic equilibrium spacing interconnected via spring-like bonds in their ground state. SEQ are fundamental, indivisible Planck-scale entities—their structure remains intact under any deformation or energy fluctuations. The sub-Planckian regime governs the spatial elasticity of the network (including elastic potential storage and release), while quantized energy transfer occurs exclusively through interactions between SEQ. Within this framework:

The spin degrees of freedom of SEQ and their elastic bonds remain decoupled, preserving independent dynamical regimes.

Under perturbation, the system responds by modifying SEQ resonant frequencies while generating compressive/tensile forces.

This elastic response is nonlinear and asymmetric.

SEQ are stable, indivisible structures composed of sub-Planckian components. SEQ' spin emerges from collective space transformations at the sub-Planck level. This ensures the spin degrees of freedom do not interfere with elastic deformations in the SEQ network. This architecture naturally protects spin dynamics from elastic disturbances.

At the sub-Planckian scale, the elastic properties of the underlying substrate impose an upper bound on the spacing modulation and tension between adjacent SEQ. This fundamental limit ensures that extreme deformations (e.g., near black hole singularities) cannot disrupt the topological integrity of the SEQ network.

In this model, the harmonic oscillation intervals of SEQ are integer multiples of Planck time(t_p). Consequently, all dynamic processes—including elastic strain

interactions, harmonic conduction, as well as scalar, spinor field transmissions and other energy conduction mode induced by rotational axis dynamics—are fundamentally constrained by the discrete Planck-time intervals. This property inherently ensures the model's consistency with the discrete-time hypothesis in quantum mechanics and quantum gravity theories.

Any discrete model of spacetime must confront the challenge of restoring spatial isotropy so as to remain compatible with the Lorentz-covariant rules established by observation. Beyond the isotropy mechanism tied to the topologically dislocated configuration discussed in §1.3, an alternative is to let the SEQ lattice spacing be sufficiently large for sub-Planckian elastic constituents—whose characteristic scale is far below the Planck length—to fill the network uniformly. Provided that, within the precision accessible to cosmological observations, the statistical distribution of these constituents yields a dispersion relation that is effectively Lorentz-covariant, macroscopic isotropy emerges naturally and remains consistent with all current observational data.

1.7 If matter truly traversed space, it would require modification of spacetime's adjacency relations. Yet black holes—despite their extreme mass—preserve local spacetime topology (as evidenced by smooth light bending). This implies that apparent particle motion must instead represent propagating excitations of spacetime itself, consistent with GR. The speed of light (c) constitutes the maximum excitation propagation rate in space.

1.8 Matter-Spacetime Unity: Matter is a part of space

General Relativity tells us that matter affects the metric of space, and the metric of space in turn determines the motion of matter along geodesics. In other words, matter and space influence each other.

Algebraic module theory and category theory further reveal that two entities capable of mutual interaction must share part of the same algebraic structure. In categorical terms, relations are morphisms and interactions are structural.

Expressed in physical language, this means that matter and space must share a common underlying basis — hence, matter and space-time are one.

1.9 Quantum-Elastic Two-Layer Fiber Bundle Model|Common Base Space Platform for General Relativity and Quantum Field Theory

This quantum-elastic base space model, where the underlying structure consists of elastic space components at sub-Planckian scales, forming a topologically homeomorphic spatial network. Within this network, structurally stable space elementary quanta (SEQ) and nodal quantized spatial networks are constructed from sub-Planckian components. The SEQ are interconnected via sub-Planckian elastic components, forming a dual-layer fiber bundle base space structure, where the sub-Planckian components serve as the foundational substrate and the space elementary quanta form the secondary layer.

On this base space, due to the fact that the space elementary quanta themselves are composed of sub-Planckian components, the spin of these quanta is essentially a topologically protected spin excitation conduction wave on the sub-Planckian topological network. This allows for the decoupling of spin and elasticity without altering the fundamental spatial topology. Matter is fundamentally a compressed excitation state of space, and the motion of matter is essentially a topologically protected excitation wave on space. The motion at each level is essentially a wave on the substrate of the lower level, thus preserving the topological homeomorphism.

Based on this structure, it becomes straightforward to define two fundamental connections: the gauge group of quantum field theory (QFT) and the metric tensor of general relativity (GR). This enables the model to simultaneously accommodate the mathematical frameworks and phenomenological interpretations of both quantum field theory and general relativity.

Between these two connections, there also exists a categorical morphism: the geometry-frequency mirror, which will be discussed in Section 13.

In this quantum-elastic spatial network model, the energy carried by space elementary quanta can represent the quantum spin-resonance energy of space. Additionally, the elastic potential energy of space can be decomposed and calculated across relevant regions of the space elementary quanta with the mirroring principle between geometric deformation and quantum resonance frequency introduced in section 13. Thus, the entire spatial energy distribution can be represented solely by the energy carried by the space elementary quanta.

Once the energy distribution across space is constructed using the energy carried by the space elementary quanta, each step in the dynamic energy distribution of this quantum-elastic space corresponds to each step of spatial transformation. The second law of thermodynamics manifests as a process of energy distribution homogenization. Based on this, multiplicative entropy can be used to define the entropy value of each spatial transformation, which can then be mapped onto the time set. This provides a clear definition of cosmic time and embeds the thermodynamic time arrow of the second law of thermodynamics into the models of general relativity and quantum field theory.